



Public goods and the evaluation of complex outcomes in space economy

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ARTICLE INFO

Keywords:

Existence value
Merit good
Private good
Public good
Space exploration
Total economic value

ABSTRACT

This paper proposes a fusion of two economic paradigms, the economic goods framework and the Total Economic Value (TEV), to structure the evaluation of complex space programs. It illustrates the dynamic relationship between different types of economic goods, including the institutional integration of manufactured private goods into public goods. Drawing on methodologies from environmental economics, particularly categories such as option and existence values, the approach addresses less obvious outcomes of space activities that nonetheless significantly shape public perceptions and judgments of program effectiveness. Current institutional developments in space activities demonstrate the relevance of economic goods and value epistemology as a framework for understanding and designing space policy.

1. Introduction

Since the first space launches in the late 1950s, the space industry has evolved into a network of actors shaping interconnected value chains that form the core structure of the space economy.

The growing dependence of global digital infrastructure on space assets [1], combined with outer space's value as both a source of human inspiration and an infinite resource for development [2], underscores the need to methodically integrate the space economy into the global economic system [3]. The conceptual foundations of this integration include measuring the size of the space economy and understanding how its outcomes are woven into the fabric of global purposeful activities.

Recent comprehensive works on measuring the global space economy include the synthesis by Jones and Weeden [4], which compiles and compares methodologies and metrics, and Crane et al. [5]. Bryce Technologies' model [6] of the space economy, featured in the Satellite Industry Association's State of the Satellite Industry Report, estimates the global space economy at approximately \$415 billion, with more than 70% attributed to commercial space services and equipment, while the remainder derives from government space budgets.

The economics of commercial space equipment and services parallel those of the international information and communication sector. Public space expenditures also generate products and services, whether contracted out or directly produced by entities like NASA, but their ultimate

outcomes differ from private-sector value chains. Turning to one of the fundamental concepts of microeconomics, the theory of economic goods, we identify most outcomes of public-sector space activities as pure or impure public goods [7], while noting that not everything produced by the public sector is a public good, and not all public goods originate from the public sector. Given that the relationship between public goods and the public sector is not one-to-one, we examine more closely the place and role of public goods within the broader context and outcomes of space activities.

Drawing the necessary distinction within the public good concept in studying the space economy involves differentiating naturally existing resources and human-made products. The 'space as commons' framework builds on Ostrom's fundamental classification [8] and has been further developed in numerous works, including a critical analysis of space-related aspects by Hertzfeld et al. [9] and a recent overview by Patton [10]. While this perspective is essential for examining the legal and economic foundations of space resource utilization [11], our focus will be on the second aspect, the classification of space-related goods as human-made rather than naturally occurring.

While considering the space economy as a whole, we pay special attention to space exploration, understood as activities oriented toward discovery rather than application¹ and funded primarily, though not exclusively, through public expenditure. As we will observe, space exploration encompasses the full range of value types, making it critical to our analysis.

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¹ The Oxford English Dictionary defines exploration as 'the action of travelling to or around an uncharted or unknown area for the purposes of discovery and gathering information' (<https://www.oed.com/search/dictionary/?scope=Entries&q=exploration>). Thus, space exploration encompasses human spaceflight and basic space research but excludes space applications.

<https://doi.org/10.1016/j.spacepol.2026.101762>

Received 21 October 2025; Received in revised form 25 February 2026; Accepted 15 March 2026

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Abbreviations

EU	European Union
GPS	Global Positioning System
ISS	International Space Station
LEO	Low Earth Orbit
NASA	National Aeronautics and Space Administration
OST	Outer Space Treaty
PGIC	Public Good Input Chain
PNT	Positioning, Navigation, Timing
SWOT	Strengths, Weaknesses, Opportunities, Threats
TEV	Total Economic Value

Some sources [12–14] treat space exploration itself as a public good, while others [15] emphasize that it generates specific types of public goods through various applications and spillovers.

The additional motivation for categorizing space industry output stems from the historical trajectories of global space programs and their resulting intersectoral developments. The increasingly privatized approach in the United States [16], along with the specific institutional dynamics in Russia [17] and, to some extent, in India [18], highlights the need for a more transparent framework to structure the goods and value generated by space activities, particularly in space exploration. Given the growing interdependence between public goal-setting and corporate implementation, a more precise delineation of goods produced and their creators is essential. A good example is the concept of making humankind a multiplanetary species [19], where corporate strategizing overlaps with the pursuit of pure public good.

We apply approaches for structuring public goods from environmental and historical preservation domains [20] to space activities, aiming to improve analytical transparency and resolution. We find the concept of the Total Economic Value [21], including non-use and option values, relevant for assessing most intangible outcomes of space exploration.

This paper contributes to conceptual/theoretical research across space policy, institutional economics, and public value theory. It builds analytical clarity by synthesizing and adapting frameworks to the evolving space sector, constructing a unified model for interpreting the generation and evaluation of space economy goods across public and private domains, rather than offering empirical data or normative policy recommendations.

The text is structured to guide the reader from a general problem setting toward a detailed analytical framework for understanding and evaluating the diverse outputs of space-related activities. Section 2 presents a selective literature review, highlighting relevant economic, institutional, and policy-oriented approaches to the recognition of economic goods and value, with a particular emphasis on the space economy.

Section 3 proposes a typology of goods produced by space missions, discusses externalities and public good characteristics, and introduces the Public Good Input Chain (PGIC). Section 4 turns to the Total Economic Value (TEV) framework and adapts it to the space context.

Section 5 discusses the implications and relative positioning of the suggested Economic Goods-Total Economic Value (EG-TEV) framework regarding the known approaches and paradigms. Concluding Section 6 reflects on the contribution, limitations, and directions for further research.

2. Literature review

Analyzing goods generation in the space economy and the associated value-for-money involves two major components. The simpler, more transparent one is associated with the provision of intended goods and

services through ordinary contractual transactions [22]. In these cases, the intended result is fully captured by the delivered product or service, and the value created is realized within the completed transaction. However, such activities may also produce externalities that extend their effects beyond direct exchange. When consumers are households or businesses, the intended value chains end here, and evaluation relies on straightforward market indicators [23]. In most cases, the state acts as both initiator and primary customer, defining objectives and financing execution through public programs. Here, the intended result extends beyond the supplied goods and services, encompassing collective outcomes such as scientific knowledge, cultural or strategic gains, or other forms of non-market value.

When contract deliverables are sold to the state as part of national space exploration or defense programs within the relevant input chains (see Subsection 3.2), the outcomes require more specialized evaluation. Accordingly, specialized value metrics are critical for outcomes that are not delivered directly through transactions between economic actors.

The usefulness of space exploration can be assessed through several distinct approaches. These approaches generally examine the generation of effects that vary in their tangibility and measurability, aiming to articulate a ‘value for money’ produced by exploration efforts [24]. Four dominant perspectives frame this value through their effects, each corresponding to a distinct type of good.

The first approach considers direct productivity. It posits that space exploration yields outcomes justifying investment. Fundamental scientific knowledge from exploration missions is a central example [25,26]. Ideally, these outcomes should align with those of other endeavors in the public good domain. For instance, scientometric efficacy, measured by the number and quality of research publications, is used to compare space research missions with comparably costly megascience projects such as the Large Hadron Collider or the International Thermonuclear Experimental Reactor [27].

The second approach deals with existential productivity. It suggests that the very conduct and intensity of space exploration by a nation or international body serves as a public good, regardless of how the results are measured. This value arises from societal perceptions: space exploration enhances national prestige, projects symbolic leadership, and contributes to broader dimensions of global competitiveness [28,29]. The existential metric is either applied independently or serves as a component of the TEV (Total Economic Value) measurement [20], which we further discuss in Section 4.

A third approach, often implicit in discussions of long-term exploration, views it as expanding the domain of future utility – effectively generating potential value by extending the scope of human activity. Crucially, this future use remains fundamentally indefinite. Space exploration is often paralleled with American expansion into the mainland, associated with the frontier concept and the broader idea of Manifest Destiny [30]. Echoing this, Donald Trump invoked ‘our manifest destiny into the stars’ in his 2025 inaugural address [31]. The frontier is frequently cited as a symbolic foundation of humanity’s innate drive to explore, expand, and establish [32]. Yet the originator of the frontier thesis, Frederick J. Turner [33], emphasized its instrumental value, writing: ‘... The people of the United States have taken their tone from the incessant expansion which *has not only been open but even forced upon them.*’ (italics added – D.P.) Hence, acknowledging the historical continuity in territorial expansion into space does not preclude the need to articulate its concrete motivations. In cases where these motivations are not associated with particular utilitarian goals, as in today’s space exploration, one might consider them to embody option value, promising benefits deferred to an indefinite future. It is worth noting that the frontier itself is no longer universally perceived as positive. Schwartz [34] argues that ‘there is much room for skepticism about the role of the space frontier with respect to cultural diversity, scientific innovation, and democratic governance.’ Moreover, these days the frontier narrative is often critically associated with colonialist thinking [35].

Finally, the external productivity approach attributes the value of

space exploration to the positive externalities it generates, such as improvements in infrastructure, education, scientific capability, and technological innovation. These second-order effects often form the practical justification for public funding, particularly when the direct utility of exploration may appear distant or uncertain [36].

These four perspectives are analytically distinct yet often overlapping; individual programs can contribute to multiple value categories.

In this article, we rely on the theory of economic goods, especially regarding public goods, to jointly analyze the products and values produced during space activities. The theory of public goods, as originally formulated by Samuelson [37] and expanded by Musgrave [38], remains foundational for understanding the government's role in providing goods that markets typically undersupply – namely, those characterized by non-rivalry and non-excludability, such as basic scientific research, free-for-all global satellite navigation service, and planetary protection [39–48]. We consider space public goods in more detail in Section 3.

In general, it is interesting to follow the different views in two major collective monographs published three decades apart. The seminal Space Economics compendium [49] mentions the concept of public good only in the context of the Landsat privatization [50] and also as part of the ‘quasi-public good’ concept when discussing the industrial spinoffs [36], while the contributors of the recent Oxford Handbook on New Space Economy [51] broadly refer to this category when discussing the role of economic sectors and the digital economy.

Although linking space activities with the concept of public good is common, the interpretation varies based on each researcher's approach, assuming they even recognize a public good category (not all do; see, for instance, Ref. [52], [p. 137] for a sharp criticism of the public good category as such). Given increasingly complex, interconnected inter-sectoral relationships [53], we argue that categorical harmonization and transparency are essential for theoretically understanding the space industry's economic output.

In the next section, we turn to economic goods, the first critical element of our framework for structuring the space economy.

3. Classifying goods in the space economy

Section 3 proposes a typology of goods produced by space missions, discusses externalities and public-good characteristics, and introduces the Public Good Input Chain (PGIC) as an analytical tool to trace how combinations of private and intermediate outputs may culminate in public value.

3.1. Conceptual and methodological foundations

A widely adopted classification of economic goods is presented by Ostrom [8,41] in the form of a quadrant chart. This matrix classifies private goods, public goods, club goods, and common-pool resources (commons) based on two attributes: consumption rivalry and excludability. A good is excludable if individuals can be prevented from consuming it. A good is rivalrous if consumption by one individual diminishes others' ability to consume it [44]. Table 1 summarizes four basic types of goods and provides the most obvious examples from the space economy domain.

Even if a public good is most often provided by the state, this classification reflects the goods' nature, not the economic sector (public or private) that produces or manages them.

The goods-based typology is helpful for analyzing key aspects of the modern digital economy [54], the organization of science [55], global development [56], and the governance of shared resources [57]. These areas are directly relevant to the analysis of space economics [58].

Applying this quadrant model to space, however, requires refinement. Specifically, we address the distinction between public goods and club goods when analyzing complex domains such as space activities,

Table 1

Quadrant of economic goods with example produced goods from the space domain.

	Rivalrous	Non-rivalrous
Excludable	Private Good Personal satellite communication Commercial Earth remote sensing Space tourism	Club Good Direct-to-home TV Differential satellite navigation
Non-excludable	Commons (not produced goods)	Public Good National security Fundamental space research Civilian navigation field

national defense, or education. Although such services are often considered public goods [7], examples such as national defense – clearly non-global – and space programs pursued for national prestige [59] suggest that their benefits are limited to specific populations (typically citizens). Pirtle [35] cites the original NASA mandate asking: ‘How should NASA work to the ‘benefit of all [humankind]’ ... Who is included in that statement and how should they be involved?’

Territorial scope of a good's applicability is not an explicit criterion in the standard rivalry-excludability matrix, though it often appears implicitly in related discussions. Beetsma [60] discusses a European defense as ‘a public good when the public of reference is the citizens of the EU.’ The idea of ‘public of reference’ in conventional public goods, even if not identified as such, has led to the development of the global public goods framework [43,61], which explicitly removes these constraints. It has also stimulated interest in regional public goods [62].

Critically, territorial limitation for public goods functions similarly to club good excludability [63]. In both cases, access to a non-rivalrous good is limited to a defined group: citizens of a nation, inhabitants of a country, residents of a geographic area (as with street lighting), or other defined populations. In the case of limited (non-global) public goods, this boundary is set primarily by physical factors (geography); in club goods, it is enforced through designed exclusion. Thus, we define limited public and club goods as non-rivalrous and non-excludable within the group limited by either physical access (for limited public) or designed exclusion (for club).²

Two similarly named yet distinct subfields illustrate different applications of public-goods logic. Planetary protection, concerned with safeguarding extraterrestrial environments from contamination following Earth spacecraft landings [64], parallels canonical environmental TEV modeling. Meanwhile, planetary defense, which involves protecting Earth from hazardous near-Earth objects, exemplifies the potential for deploying a previously unseen form of pure public good with defense characteristics [65].

There is also a category of merit goods. We follow Fiorito and Kolintzas's [66] definition: merit goods are goods ‘provided by the government on paternalistic grounds (e.g., individuals ought to consume them even if they (could) would not, acting on their own self-interest).’ Here, merit goods are an intentionally produced subset of the full continuum of public goods, including government-owned space infrastructure and science research data.

On this basis, we further consider two types of space-related public goods, including territorially limited public goods and club goods, which may both be regarded as impure public goods in this context and differ in their production value chains. The first type, *merit public goods*, includes public goods that are target outcomes of contracts and programs, most often government ones. The second type, *spillover public goods*,

² We do not delve into the subtle issues of limited public and club good differentiation for more complicated cases – for instance, when citizenship is a limiting factor – and recommend insightful work [63] to those interested.

includes public goods generated as externalities, the indirect results of other goods' production. An example is the general stimulation of national economic activity resulting from space industry operations.

While originally linked to addressing market failures [67], recent perspectives emphasize a proactive role of the 'entrepreneurial state' in producing certain goods and assuming risks associated with innovation [68]. Given the vast scope and lack of consensus regarding the state's role, this paper focuses on the constructive aspects of goods generation in the space sector, drawing on empirical observations of existing projects and programs rather than analyzing sectoral dynamics.

The goods structure specifies what is produced; value frameworks specify the purposes of production [69]. The value of produced private goods is relatively straightforward, typically defined by market transactions. Evaluating public goods, both pure and impure, is more complex. Two well-established categories of public goods are national defense and fundamental science. Although less tangible than satellite application-related private goods, both have a long tradition of value-for-money assessment, grounded in the decades-long (if not centuries-long) methodological base of public finance [70,71].

However, the value of public space activities extends beyond defense, basic research, and industrial externalities. There is more to be considered. Scholars propose various paradigms for holistically evaluating space exploration beyond traditional metrics. Baum [72] introduces ethical dimensions, while Lawrence et al. [73] incorporate considerations of space environmentalism. In this paper, we propose a framework to discuss the full continuum of values delivered to different beneficiaries by goods generated through space activities, with particular emphasis on the 'hidden' value of space exploration that traditional paradigms do not capture.

3.2. Institutional integration of merit public goods

Private and public goods differ in consumption: non-rivalry and non-excludability (the latter applies only to public goods) define them. Despite these differences, both private and public goods must be produced first. Production involves both the physical deployment of infrastructure and the creation of an institutional environment governing consumption through specific rivalry and excludability attributes. In the space economy, a good's nature is shaped by consumption modifications at the value chain's end. Complex, multilayered technical and organizational systems in Earth observation, satellite communication, or global navigation may yield component public goods that serve national defense or commercial markets, depending on how procurement and distribution are fine-tuned [74].

The physical deployment and operation of infrastructure producing public or private goods do not fundamentally differ, especially given the growing role of public-private partnerships and Anything-as-a-Service models [75]. Typically, the industry produces goods and services that are private goods by nature. In the most common case of merit good production, governments contract with industry to produce these goods/services, add public-sector value (e.g., via state-owned enterprises or military organizations), and convert the result into public goods (pure or impure).

We introduce a new category: the Public Good Input Chain (PGIC), a sequence of intermediate goods, services, and resources required to produce a public good, excluding the final institutional integration phase that converts these inputs into the publicly accessible, non-excludable final good. It encompasses all material, technological, and human capital essential for creating the public good.

The PGIC is instrumental in clarifying the separation between the tangible production of public goods and the institutional frameworks that govern their consumption. Traditional economic models often blur this line, making valuation difficult. We use the PGIC framework further in discussions of value chain structures in the space economy.

While not ontologically novel, PGIC's utility lies in differentiating the production segment of the public good's value chain from the

institutional phase that establishes the consumption mode, thereby defining whether the created good is private, public, or club.

The PGIC highlights elements directly involved in production, such as infrastructure, technology, labor, and raw materials, before incorporating them into policy/governance frameworks that make the good available to society. Private firms or state-owned enterprises typically produce these intermediate goods; the final public good benefits all individuals within the 'public of reference.'

The reusable Crew Dragon spacecraft demonstrates how a single production and operational chain can serve different institutional contexts. The Public Good Input Chain, covering design, manufacture, integration, and launch, remains technically uniform across missions, though individual procedures and objectives vary.

Under NASA's Commercial Crew Program [76], Crew Dragon flights support ISS operations within a government-led PGIC that sustains collective human-spaceflight infrastructure. Within this setting, commercial activities such as Space Tango's biotechnology work, Redwire's manufacturing facilities, and the iBOSS modular system illustrate how private actors function inside a publicly managed research framework through the National Laboratory mechanism.

Privately financed missions such as Inspiration4 [77] and Polaris Dawn [78] use the same PGIC but combine private outcomes – corporate visibility, technological validation, and mission participation – with deliberately public components such as scientific and biomedical research or philanthropic fundraising supporting merit- or public-good-oriented activities. Educational and outreach effects are treated here as co-produced externalities, positive spillovers accompanying public communication and engagement.

The PGIC concept, therefore, shows that technically similar supply chains can yield different mixes of intentionally produced goods, depending on institutional integration, the 'last link' connecting production to its final public or private outcomes.

3.3. Externalities and spillover public goods

Spillover public goods arise as side effects (externalities) of public or commercial programs.

The GPS satellite navigation system exemplifies the transition from a club good to a public good. Initially a military club good providing positioning, navigation, and timing (PNT) signals, GPS gained public good status after Reagan permitted commercial airline use (1983) and Clinton stopped degrading civilian accuracy (2000) [79]. GPS as a public good might be considered a positive externality resulting from the military system operations.

Benefits (e.g., free navigation) spill over to non-contracting parties (global population, businesses) who do not pay for the development and operations. This unaccounted benefit is an externality, an unintended side effect of the primary contract purpose (national defense), additionally adopted for civilian/commercial use without compensating original stakeholders.

Internalizing this externality may involve licensing agreements or user fees, public-private partnerships to share costs and benefits, or taxes or fees on companies heavily relying on the publicly built system for commercial operations.

3.4. Typology of outputs in the space economy

Fig. 1 illustrates the genesis of major types of goods in the space economy, originating from 'physically manufactured' core private goods. Beyond merit public goods, we include two types of spillover goods. Both arise during the manufacture of core private goods. Spillover public/club goods – accessible beyond core private good customers and contractors – emerge as positive externalities [63]. In contrast, spillover private goods relate to industry development as a side effect of producing core private goods [36].

A special category is platform-subsidized quasi-private goods. These

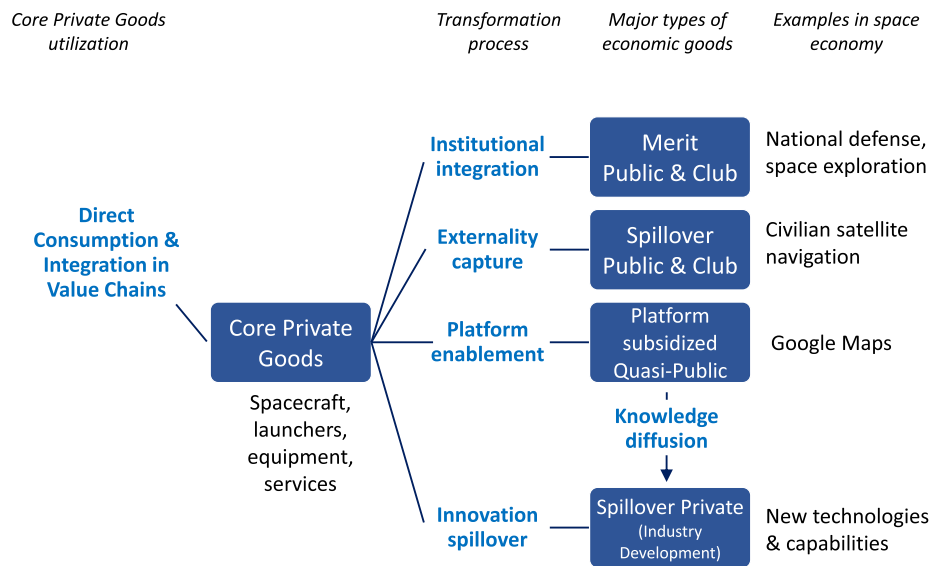


Fig. 1. Economic goods in the space economy.

services are offered free or at low cost via a platform, are indirectly financed (e.g., through advertising or data licensing), and foster broad ecosystems. Google Maps, closely related to Earth remote sensing, is a well-known example. We include this category due to its growing prominence and its classificatory complexity within the traditional public/private framework.

The next section considers values and valuation and introduces our suggestion of adopting the Total Economic Value framework. At the end of Section 4, we use the presented goods structure for cross-referencing with values.

4. Valuation through the Total Economic Value (TEV)

Section 4 turns to the Total Economic Value, adapting it to the space context and expanding it to capture both direct and indirect values, including option, heritage, existence, and bequest values.

4.1. Applying TEV to space economy goods: EG-TEV framework

Beyond tangible outcomes such as commercial or military use, space endeavors produce scientific knowledge, indirect effects (both internal and external, as noted by Bach et al. [36] and others), and complex public goods. These public goods range from national pride to comprehending humankind's unlimited developmental future in space.

This value arises from goods generated either as externalities of specific projects or intended outcomes of activities, most often by the public sector, which take the form of impure public goods limited by access (Section 3). Despite frequent references to education, inspiration, and intangible benefits linked to space programs, these elements remain conceptually underdeveloped.

Comprehensive identification and evaluation of indirect value are crucial: they underpin the continued legitimacy of the public sector as a core actor in space exploration. A robust framework, tailored to the space domain, that accommodates non-market effects is needed, but it must remain compatible with the evaluative methodologies used in other policy fields.

Given the acknowledged importance of the existential dimension of the space economy [2, p. 1], along with other forms of indirect value previously discussed, we use the Total Economic Value framework as a structured articulation of socially recognized but analytically under-

specified dimensions of value.

Fig. 2 summarizes the conceptual relationships among direct and indirect components of TEV relevant to space activities; it does not imply a quantitative aggregation procedure, which would require a separate methodological framework.

TEV accounts for tangible benefits from the direct and indirect use of goods and services, as well as intangible benefits that exist regardless of direct consumer engagement [21,80]. Traditionally applied to environmental considerations in cost-benefit analyses of development projects [81], it also has the potential to evaluate other complex socioeconomic phenomena.

It provides a structured framework for assessing the full range of value generated by goods and services, especially when traditional market valuation fails to capture their broader societal impact. TEV distinguishes **use values** (benefits derived from actual or potential use) from **non-use values** (importance of a good or service independent of its consumption).

Use values are typically divided into two categories. **Direct use values** arise from immediate interactions (e.g., utility gained from using a service, product, or public infrastructure). **Indirect use values** capture secondary benefits (e.g., environmental, educational, or systemic

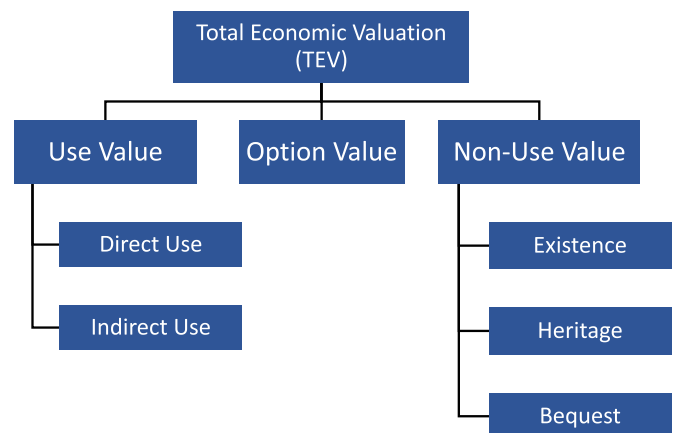


Fig. 2. The Total Economic Value framework (compiled from Refs. [20,21]).

advantages from a good's existence).

Option value is the value of preserving the possibility of future use.

Non-use values are independent of use: **existence value** (traditionally associated with awareness that something exists), **bequest value** (leaving it for future generations), **heritage value** (its relevance to present identity or continuity). Together, these categories ensure that the TEV framework encompasses both measurable utility and more abstract, but socially relevant, sources of value.

A novel aspect is the recognition that goods that give rise to all types of value covered by the framework can be purposefully generated, rather than treated as given inputs. Thus, their value not only justifies efforts to preserve them but also supports investing in their creation.

Applying this framework provides a holistic perspective on the value created by space-related activities. This structured approach is crucial for understanding consumption patterns and evaluating long-term economic impacts beyond immediate commercial outcomes.

The use values of private and public goods are a practical category for assessing the commercial component of the space economy. In contrast, option and non-use values capture the broader societal and economic effects of an asset or service, regardless of whether individuals personally utilize them.

Applying the TEV to created goods entails significant challenges. Created goods, such as space infrastructure and technological systems, differ fundamentally from naturally existing resources: they are intentionally designed, dynamically evolving, and embedded in complex institutional and value chains. These distinctions create the need to reconsider how traditional TEV components—use, option, and non-use values—are defined and measured in this context.

The EG-TEV framework identifies the full spectrum of potential value categories rather than their quantitative measure and associates them with the economic good taxonomy. In practical decision-making, these values must be considered relative to the costs, risks, and opportunity constraints associated with each alternative, but the cost–benefit comparison itself lies outside the present conceptual focus. EG-TEV fuses the economic goods taxonomy with Total Economic Value structuring to establish a unified analytical framework that links categories of created goods to their associated value dimensions.

The following subsection illustrates this by focusing on option value, examining how temporal uncertainty and long investment horizons shape value realization for created goods.

4.2. Option value in created goods: temporal framing

Option value [20] reflects the willingness to invest in maintaining access for potential future use, despite the lack of immediate benefits. For created goods, such as advanced infrastructure or emerging technologies, the temporal dimension is especially salient. Often, the expected payback period for these goods extends beyond conventional planning or forecast horizons, challenging standard investment appraisal methods.

Comparing hypothetical extraterrestrial infrastructures illustrates this. A lunar base mining Helium-3 [82], once a popular candidate for thermonuclear fuel, exhibits use-type characteristics, with foreseeable and reasonably planned payback. This enables private investment via commercial returns and discounted cash flow valuation. Nonetheless, substantial risks remain – technological, market, and regulatory uncertainties – that continue to challenge investors despite the clearer economic outlook. In contrast, a Mars colony [83], as seen today, embodies option value: its potential benefits, such as terraforming or resource exploitation, are distant and fundamentally uncertain. These features place the Mars settlement beyond typical private investment horizons, necessitating sustained public funding justified by long-term societal interests and intergenerational equity.

Evaluating option value thus demands frameworks incorporating temporal uncertainty, irreversibility, and institutional capacity to manage long-term commitments. This temporal and institutional

perspective is essential for adapting TEV methodologies to capture the unique valuation challenges posed by created goods with extended and unpredictable benefit horizons.

China's strategic planning document [84] exemplifies this. It integrates diverse exploration fields, such as aerospace, deep sea, and deep Earth, under a unified 'Future Space' framework, treating them as a single spectrum distinguished by differing time horizons. Such integration aligns with the concept of option value, which emphasizes the importance of maintaining access to uncertain future opportunities across domains with variable and often extended temporal profiles.

4.3. Non-use values: intergenerational connection

Non-use values are traditionally subdivided into existence, bequest, and heritage values. Among these, the existential dimension of space exploration is the most frequently discussed, often associated with grand visions such as space colonization, transhumanism, and terraforming [85]. However, the TEV interpretation is more restrained: while popular discourse links existential meaning/value with survival and continuity of humankind and civilization, the canonical TEV framework assigns existence value to goods whose worth lies in the mere knowledge of their existence.

In line with environmental economics literature, we treat existence value as one component of non-use value within the Total Economic Value framework, where non-use values are characterized by independence from direct utilization. By contrast, existential value refers to broader symbolic, identity-related, and civilizational meanings associated with space exploration. In environmental contexts, existence value reflects awareness that a site/object exists – even if inaccessible for any form of use, including aesthetic appreciation. Similarly, celestial objects like Saturn's rings hold this value. Yet a crucial distinction arises: the environmental TEV model primarily aims to justify conservation efforts against anthropogenic destruction, whereas the proposed space exploration EG-TEV framework seeks to encompass the full array of benefits and values that humanity associates with actively pursued extraterrestrial endeavors and goods created in this way.

We argue space exploration embodies distinct locational existence value based on 'awareness of accessibility' rather than solely 'awareness of existence.' Its worth is framed as 'It is good that it is accessible to humankind' rather than 'It is good that it exists.' This emphasis on accessibility complements the traditional notion of existence, serving a similar functional role in the valuation of space exploration, despite diverging from the classical definition of existence value.

Bequest and heritage values complement this, intersecting with concepts like the common heritage of humankind [86]. Bequest value transmits non-use values (e.g., national pride and social identity) to future generations; heritage value reflects accomplishments and cultural identity inherited from ancestors.

Non-use goods encapsulate critical 'intangible social affection' underpinning space programs. Launius [87] summarizes multiple surveys: 'while Americans may not know much about the space program, they have a largely favorable opinion of it—over 70 percent say they have a favorable impression ... The American public has long held generally positive attitudes toward the space program, but is not very familiar with its details.' Mahoney [88] notes that 'for most people, space exploration is, and will forever remain, primarily an emotional pursuit.'

Another example is sustained Russian human spaceflight through the politically and economically challenging 1990s and 2000s [89]. A recent survey [90] reveals enduring public interest in space exploration and its specific facets. More broadly, values tied to intergenerational continuity hold special significance in cultures where preserving social values and achievements across generations forms a cornerstone of national identity and policy frameworks [91].

This conceptual exercise connects value types to underlying goods via temporal distinctions. It differentiates use, option, and non-use values for created goods by time-to-utilization profiles: use value

corresponds to goods with predictable, though possibly variable, timing of benefit realization; option value involves goods whose time to value generation is fundamentally uncertain or depends on unforeseeable developments (such as ‘black swan’ events [92]); and non-use value pertains to goods whose benefits are not associated with utilization as such and therefore lack a definable time of commencement. This demonstration serves as a foundation for further refinement in adapting TEV methodologies to created goods.

4.4. Cross-referencing goods and value categories

This subsection attempts to map economic goods produced by space activities to core components of the Total Economic Value framework. Table 2 is structured with rows corresponding to the primary categories of generated goods (see Subsection 3.4) and columns representing the principal elements of the TEV framework (see Subsection 4.1).

Evaluation applies to specific goods at a particular value-chain tier. Since it reflects consumer value, evaluation depends on the consumer type and the utility derived from the good. The same good holds different values for different consumers. For instance, a communication satellite holds different values for its manufacturer, operator, and user – contract revenue, operational efficiency, and service utility, respectively – illustrating that evaluation depends on consumer type and context. This contrasts with environmental TEV, which evaluates natural objects holistically for all users and ‘no-user’ beneficiaries.

Most physical outputs are core private goods associated with direct use value, whereas merit public or club goods are intentionally developed to serve collective needs. Distinguishing between direct and indirect use values often depends on the system's architecture and purpose.

The widely held notion that ‘space exploration has existential significance’ can be formalized as a spillover public good – ‘Awareness and Recognition of Extending Human Knowing and Presence in Space’ – which, lacking contractual deliverables yet being non-excludable and non-rivalrous, fulfills the criteria of a public good. Within the TEV framework, its existence value is a non-use value derived from mere knowledge of this collective endeavor. Analogous spillover goods arise from public perception of corporate or technological contributions to exploration.

We also include a non-obvious pair of goods related to extraterrestrial resources to illustrate the suggested differentiation between option and bequest values for created goods. Option value applies when future utilization is feasible under current institutional and legal frameworks, regardless of technological or economic hurdles that may evolve independently. Bequest value applies when institutional or legal frameworks currently preclude utilization, preserving the good for future generations who may benefit once those frameworks change. For example, ‘Intellectual property (IP) and technologies for space resource excavation’ may align with either option or bequest value, depending on the interpretation of the Outer Space Treaty (OST) [93]. Under a restrictive reading that precludes commercial exploitation, such goods represent bequest value – preserved for future use once frameworks change. Under a permissive interpretation, they represent option value – future benefits feasible within existing law.

This mapping provides a basis for developing evaluation models to rank goods, especially those with non-use values, and potentially to associate them with quantitative indicators within the EG-TEV framework. Although preliminary, this represents, to our knowledge, a first attempt to apply TEV systematically to the space economy and its outputs, and it may serve as a foundation for further multidimensional ‘value-for-money’ analyses in space exploration.

4.5. Measuring challenge

Associating goods with value is a necessary step toward developing detailed valuation methods that enable comparisons across space projects and with initiatives in other domains, such as fundamental science,

Table 2

Goods and value categories in the space economy.

Good Types	Use values		Option values	Non-use values		
	DU	IU		EX	HE	BQ
Core Private						
Spacecraft, systems, equipment & constellations	X					
Enabling infrastructure and data ecosystems supporting multiple applications	X					
Commercial satellite services	X					
Third-party applications	X					
Human capital	X					
Emerging technologies		X				
IP and technologies for space resources excavation (permissive OST)			X			
IP and technologies for space resources excavation (restrictive OST)						X
Spillover Private						
Pilot deployments and prototypes enabling near-term commercial use	X					
New technologies and capabilities	X					
Emerging applications		X				
Collateral IP			X			
Knowledge transfer			X			
Merit Public (incl. territorially limited)						
Government civilian & military space infrastructure	X					
Earth observation & navigation services with public access	X					
Open-access data platforms	X					
Scientific knowledge	X					
Global space situational awareness		X				
National security assurance		X				
Space-environment sustainability assurance		X				
Weather-forecasting capability		X				
Public education and outreach		X				
Spillover Public (incl. territorially limited)						
Incidental environmental monitoring data	X					
Indirect benefits from public-private partnerships		X				
Stimulation of economic activity		X				
Awareness and recognition of extending human knowing of and presence in space				X		
Awareness and recognition of outer space as an infinite potential resource for humankind				X		
Entrepreneurial inspiration to increase involvement in space activities				X		
Awareness and recognition of national input into the historical achievements in space activities					X	
Awareness and recognition of transferring the national achievements in space activities to future generations						X
Merit Club						
Licensed technical standards actively used within membership groups	X					
Collaborative development platforms with controlled access		X				
Spillover Club						
Spillovers from club innovation		X				
Awareness and recognition of corporate (group) input in space activities				X		
Platform Subsidized						
Free-access digital platforms		X				
Citizen science contribution		X				

DU – direct use; IU – indirect use; EX – existence; HE – heritage; BQ – bequest.

environmental protection, or cultural heritage.

For **direct-use** private goods, valuation follows traditional market-based approaches, relying on *ex ante* pricing and supply-demand dynamics.

Option value differs from the anticipated future value primarily in its temporal uncertainty. While future profits are typically discounted over time [94], the TEV framework captures the subjective value of potential use at an indefinite future moment (see Ref. [80] for an overview). Beyond contingent valuation, option value can be modeled by treating the discounting period as a stochastic parameter. The state usually assumes funding responsibility for programs with such indefinite horizons [95].

Non-use values are usually assessed through expert-based approaches such as Delphi surveys and benefit transfer [96], or contingent valuation methods, including willingness-to-pay and stakeholder consultations [97]. Survey techniques are popular for assessing existence values [87,90,98], but they present specific challenges in the space domain.

First, most TEV studies evaluate naturally existing public goods: landscapes, ecosystems, or heritage sites, rather than intentionally created non-use goods [69]. Second, space infrastructure costs far exceed those of preserving terrestrial or heritage assets in an object-to-object comparison, complicating direct stakeholder surveys, especially when beneficiaries are the national or global population [99]. Finally, generating new non-use goods is inherently more complex than maintaining existing ones. Although secondary, existence value remains notable in human spaceflight programs, where social preference studies confirm persistent public support [19,90,100].

At present, comprehensive valuation methods primarily legitimize option and non-use values rather than provide operational tools. Before quantification, recognizing these intangible benefits in planning frameworks is critical [101]; qualitative expert ranking may suffice initially, while developing quantitative models remains a demanding next step [102].

We conclude this initial examination of TEV adaptation for the space economy and proceed to the discussion of the institutional role of the proposed constructions.

5. Discussion

This paper suggests two additions for the space economy toolbox. The Public Good Input Chain, PGIC, is basically a subset of the full public good supply chain, integrating the 'material' industry's input and excluding the public sector contribution affecting the access to the resulting good, and thus identifying it as pure or impure public good. EG-TEV is a concept that integrates the economic goods paradigm and categorization, using Total Economic Value metrics, thereby extending the value associated with space activities to account for non-direct use parameters.

Neither of these two aims to replace or challenge existing methodologies of space activities' institutional design, be it government procurement methodology for public goods and services or identification of space exploration priorities through the academic community reviewing [103] and top-level political decision-making [104]. On the contrary, we suggest these tools as an integrating link between otherwise separated intellectual domains.

PGIC is not the next big thing in institutional design. Rather, this is a small utility that provides more precise optics for observing and promoting public-private interplay. It is of universal character, and its special relevance to the space industry is defined more by the still significant role of the public sector in space than by any feature of space activities per se.

Traditional institutional theory focuses on governance structures [41] and property rights arrangements [105]. PGIC does not challenge these foundations but rather isolates a specific aspect: the material production chain that precedes institutional access decisions. Where

institutional economics asks 'how should we govern?', PGIC asks 'what is being materially produced before governance determines who can access it?'

The combination of economic good taxonomy and the Total Economic Value framework (EG-TEV), in turn, has the potential to grow into a large and complex construction. TEV appears well suited to addressing shortcomings in goal formulation, if we can neither calculate the outcomes nor even finalize the taxonomy of tangible (and thus calculable) outcomes, which is a case for space exploration beyond the satellite applications and contributing to fundamental science.

EG-TEV is different from both regular outcome evaluation frameworks and the existing environmental and heritage preservation TEV paradigm. Unlike 'traditional' TEV, it applies the full set of values, other than those of direct use, to 'made,' intentionally produced economic goods. Unlike outcome evaluation, EG-TEV extends the project's and program's output value to non-use components that remain important in the broader context.

The four value perspectives identified in Section 2 (direct productivity, existential productivity, future utility, and external productivity) represent fragmented approaches. EG-TEV integrates these as direct-use, existence, option, and indirect-use values within a unified evaluative framework.

We extend TEV beyond its established domain of natural/heritage resources to purposeful missions and human-created infrastructure of space activities. This extension is justified by the space activities' character as frontier domain exploration. Space activities are a category of human endeavor with few terrestrial parallels beyond the Antarctic and the ocean depths. The present EG-TEV approach represents a novel application whose extension beyond space activities was not considered in this research.

Conventional metrics such as key performance indicators (KPIs) and commercial project outcomes (see a classical Project Management Body of Knowledge [106] for a good overview) create overly narrow evaluative frameworks for measuring large-scale endeavors.

While frameworks such as the Triple Bottom Line [107] recognize the need to evaluate beyond conventional financial metrics, they struggle with operationalization due to incommensurability across dimensions [108]. In contrast, TEV literature demonstrates well-developed instrumentation based on qualitative content analysis, including interviews and archival data (a detailed application to circular municipal wastewater management is provided in [109]). The EG-TEV extension leverages this established methodological foundation while adapting it to the unique characteristics of space public goods.

In contrast to the narrower frameworks and those of limited practical usability, the TEV framework is inherently capacious (perhaps even exhaustively so), providing extensive parametric space for assessing less tangible and sometimes wholly intangible yet critically important dimensions. This comprehensiveness ensures applicability across diverse programs despite varying value emphases. Juxtaposing the TEV construction with the carefully built system of project outcomes in the format of economic goods (which is precisely what EG-TEV does) allows limiting the utilization of the complex and rather alien qualitative metrics only where they are necessary in terms of the nature of the produced goods.

The main challenge of TEV is operationalizing it, that is, converting non-use metrics into a set of quantitative or ordinal indicators. Existing TEV practices [109,110] usually refer to different ways of operationalizing public opinion regarding landscapes, natural objects, or artificial objects under consideration. We propose to draw a parallel with the existing practice of identifying the priorities of fundamental space research by addressing the scientific community through the decadal surveys' mechanism [103] and point to the possibility of organizing the EG-TEV-based evaluation along a similar organizational and representative structure.

We summarize the discussion of the EG-TEV framework with the SWOT (Table 3), a standard single-entity evaluation tool.

Table 3
EG-TEV framework SWOT.

Strengths	Weaknesses
Explication of the indirect use and non-use value dimensions of the space programs that were previously implicit Extending the total economic valuation towards goods to be created rather than those that preexisted Integration of two social science paradigms (economic goods and TEV) leading to methodological synergy Broadening the scope of theoretical discussion by means of adding new types of value associated with different goods created	Vagueness of the non-use value category, if applied to created goods Value assessments contingent on uncertain future institutional arrangements (cf. OST & space resources) Hurdles of measurement and comparing less 'material' types of value between project alternatives
Opportunities	Threats
Framing the survey and analysis of citizens' preferences Integration into agency decision-making processes Framework for international cooperation and global development discussions Input to decadal survey prioritization methodology Template for developing evaluation frameworks in emerging technology sectors	Premature quantification without proper operationalization Cherry-picking metrics to justify predetermined decisions Loss of clarity and interpretability in space policy decision-making

In brief, our approach provides greater clarity in juxtaposing activities, actors, and results when discussing the complex phenomenon of space activities. Returning to the economic goods paradigm, we identify the typical roles of public and private actors through the PGIC mechanism, thereby capturing the private-good component of the public-good value chain. Further, building on the created public good, we propose a specific optic of non-use value and other elements drawn primarily from environmental and cultural heritage economics through the EG-TEV framework. Given the traditions of non-use value identification, including through dialogue with communities and society as a whole [111], this enables new approaches to building a rationale and evaluation metrics for directions of space activities where not only calculation but even prediction of material output figures is complex.

6. Conclusion

6.1. Contribution to theory

To summarize the preceding analysis, we highlight the following key contributions that advance the understanding and development of space-economy scholarship and governance doctrine. These include:

Enhancing the applicability of the classical goods framework to space economics.

This includes explicitly accounting for territorial limitations that function analogously to excludability in club goods, as well as introducing the concept of 'goods created,' highlighting that many space-related public goods are actively produced outputs rather than naturally existing resources.

Introducing the Public Good Input Chain (PGIC) as a conceptual tool separating material production from institutional access arrangements.

This concept clarifies the analytical boundaries between production-chain analysis and institutional governance analysis and offers a framework for understanding public-private production boundaries prior to governance decisions. Together, these two elements improve analytical clarity and provide a more symmetrical and realistic representation of space-economy goods.

Developing an integrated EG-TEV framework that juxtaposes economic goods taxonomy with the Total Economic Value paradigm.

EG-TEV extends the Total Economic Value framework beyond natural and heritage resources to purposefully created goods. It provides a unified evaluative tool for fragmented value perspectives (direct productivity, existential productivity, future utility, external productivity) and broadens theoretical discussion by adding value types associated with different goods created through space activities.

6.2. Contribution to practice

PGIC and EG-TEV have several analytical and policy-level applications that can be summarized across five functional domains.

6.2.1. Applied research and analytics

Analyzing the production of various economic goods in space activities aligns with contemporary research paradigms in public-choice theory [112] and welfare economics [113], thereby expanding the epistemological foundation of space economics. Using the two proposed frameworks provides an additional lens for analyzing new projects in the public and private domains, enabling a clearer understanding of stakeholders' interests, incentives, and investment directions. Together, these analytical perspectives form the foundation for applied domains such as evaluation, communication, and strategy.

6.2.2. Evaluation and budgeting

The EG-TEV-based approach enables evaluation of space-exploration programs in a broader context and the justification of their value across a wider spectrum of societal impacts, thereby structuring the argument for setting priorities and allocating resources. This is particularly useful when discussing high-level preferences for various areas of non-commercial, non-market activities in the formulation of public budgets.

The fusion of the economic goods taxonomy and TEV identifies the full spectrum of values generated by space activities, including intangible and non-use values. This provides a foundation for further development of evaluation systems for space-related (both pure and impure) public goods as part of state-level strategic planning and procurement.

6.2.3. Communication and public perception

The EG-TEV framework can also be used to structure public discourse, helping to separate symbolic and existential values from market-based arguments. This framework helps shape more balanced communication strategies between agencies, businesses, and society by linking measurable results to existential values, which is especially important when discussing missions with a strong symbolic or inspirational component.

EG-TEV enriches the decision-support toolbox in space governance with attitude-measurement tools, enabling broader accounting for citizens' and communities' opinions.

6.2.4. Strategy and policy

A focused look at the common features and differing characteristics of generating various types of goods in the public and private sectors is particularly relevant in the context of the private sector's ongoing expansion into domains of space activity previously reserved exclusively for governments – including space exploration [114], habitable LEO infrastructure [115], and the commercialization of various forms of space situational awareness [116]. A PGIC-focused approach helps project architects identify intersections between public and private interests. This is particularly useful when designing architectures of shared-value creation in public-private partnership projects such as IRIS² [117], where a common supply chain supporting the creation of physical and organizational infrastructure eventually branches to generate value for distinct commercial and public users.

6.2.5. Sustainability and forecasting

By incorporating a wide range of temporally disaggregated values, EG-TEV provides a stronger foundation for discussions of sustainable

development and frames future-oriented work, thereby exceeding the temporal boundaries of qualitative outcomes analysis.

6.3. Limitations of the study

Limitations of the suggested institutional frameworks were considered in Section 5, particularly in the SWOT overview. In this subsection, we focus on the limitations of this particular study.

This is a purely conceptual study. While we used examples and cases from the literature to argue the correctness of fundamental propositions, we did not test the framework empirically. We consider this the key direction of future research.

We intentionally focused on the value elements associated with space exploration; that is, we applied the framework analysis to space applications only superficially. Extending this vision of a full value structure, including indirect use and non-use value, to directions of space activities and elements of space infrastructure that affect the digital economy and international security would require dedicated research grounded in fundamental insights into space security dynamics [118,119], and the transformative potential of digital technologies [120,121] as they intersect with space-based infrastructure.

We acknowledge the lack of a quantitative approach to non-use evaluation (see Subsection 4.5) but consider this a general limitation of the approach that will be a direction for future research in the domain rather than a shortcoming of the current paper.

6.4. Avenues for future research

Future research on the PGIC and EG-TEV frameworks can be divided into two directions.

The first direction concerns eliminating identified deficits and operationalizing the conceptual results. The second one includes the potential extension and broadening of the institutional research agenda on the space economy, based on the suggested ‘coordinate space’ of economic goods, supply chains, and values.

To improve the results, we consider that further research on survey methodology and the non-use values reflected in citizens’ opinions would be beneficial. Next, an in-depth study of the measurement and quantification of non-use and indirect-use values would address key issues in operationalizing these values.

To improve coverage, we see potential to extend the research to satellite applications and to consider the different value categories of space program outcomes that affect the global digital economy. The much stronger ‘value for money’ character of the space applications compared with space exploration leaves less room for the non-use value component. However, research on the indefinitely remote impact of satellite services and applications might call for developing corresponding comprehensive frameworks that might also be based on EG-TEV or similar systems, including ‘limited use or non-use’ components.

A potential extension of this research would be to more explicitly integrate both positive and negative externalities into a comprehensive system of outcome evaluation.

Finally, institutional research on economic goods, supply chains, and value dimensions would be instrumental in supporting the most recent developments in the field of the space economy. The recent announcement by former Google CEO Eric Schmidt and his wife Wendy of plans to privately invest in four telescope projects, including the space-based Lazuli mission [122], challenges the dominant role of the public sector in producing space exploration-related public goods. While it remains unclear whether the project will generate observation data as a public good available to research communities worldwide, or whether it will assume some commercialization of new distant-space observation capabilities, this appears to be a significant milestone in charity-based endeavors that claim a growing share of public-good generation in the space field. The news of the US DoD investing in the national industrial base by becoming an anchor investor in L3Harris Technologies’ missile

business for as much as \$1 billion [123] also calls into question some institutional fundamentals. The US public sector’s restrained approach to defense industry ownership, beyond limited control of the Organic Industrial Base [124], has long been a key thesis in discussions of differences in international space industry organization, as opposed to partial state ownership in Europe, state corporations in Russia, and state-owned enterprises in China. This now appears to be changing.

In conclusion, we emphasize that this work represents an initial effort to integrate economic goods theory and valuation approaches in addressing contemporary challenges in the space economy. Further development and practical adaptation of the proposed directions and approaches will require intensive methodological and empirical research. We consider our goal achieved if this work stimulates further dialogue and methodological development among colleagues in both industry and academia.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT and Claude AI Assistants in order to check grammar, syntax, and the structuring of the narrative. After using this service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dmitry Payson reports financial support was provided by Dresden Senior Fellowship. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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